

Hamrin Lake monitoring and sustainability in Iraq using GIS-based machine learning and remote sensing

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Abstract

The present research is focused on the study of the changes that have occurred in Hamrin Lake and to test the ability of Geographic Information System (GIS)- based Support Vector Machine (SVM) to provide reliable water monitoring. The main objective of this research is to explore the evolution of the Lake by using GIS, Sentinel-2 satellite images, and a Machine Learning ML approach, SVM classification. Based on three multi-temporal images of the years 2017, 2022, and 2026, we assessed the changes in lake extent and its surroundings. The classification results indicated that in October 2017, the Hamrin Lake area was 169.09 km², but showed a significant drop in April 2022 to 30.08 km², and a rise to almost 185.77 km² in January 2026. Such variations are driven by the drying conditions experienced within the catchment, hydrological variation, evaporation, and water management within the study area. The interpretation and consistency of the classified water boundaries with the classification in the generated maps showed that the SVM classification had an estimated overall accuracy of about 94%. The observed fluctuations highlight the need for monitoring support for sustainable water resource management and putting in place effective water allocation and conservation strategies in Iraq.

1. Introduction

The interactions of land use and land cover, vegetation, and climate with the lake's hydrology control the lake's ecosystem response, which in turn dictates the environmental sustainability of lakes [1]. These parameters can be used to track trends over time, and significant trends with possible implications of degradation or recovery of the ecosystem can be identified [2]. Climate change, population growth, upstream damming, and over-exploitation of limited water resources have resulted in high environmental stress on many of the water bodies in Iraq [3-5]. Thus, advanced monitoring approaches, which are based on GIS, RS, and ML, are important for environmental planning and sustainable development strategies [6-8]. These technologies can be used by government and researchers to identify sensitive regions,

determine the condition of these ecosystems, and apply management strategies for the ecosystem that are appropriate [9]. RS and GIS are now becoming effective tools to deliver highly accurate and rapid spatial information in a large spatial region, and are useful in environmental monitoring and sustainability analysis over large areas [10, 11]. GIS helps to integrate and visualize spatial data and analyse it, while satellite imagery is used to monitor land and water resources [12]. These technologies allow lake changes to be studied at both temporal and spatial scales, identify degradation of the environment, and provide support for sustainable decision-making processes [13]. Specifically, multispectral satellite information from the Sentinel-2 and Landsat missions is useful in terms of spatial and temporal resolution for water body mapping, vegetation cover, soil information, and land-use change [14]. In recent years, ML techniques have greatly enhanced the capability of RS analysis, increasing the success of classification and automatic feature extraction processes [15]. In recent years, the use of remote sensing, GIS, and machine learning in monitoring water resources in Iraq has been shown to be effective, especially in a climate of growing climate variability and water scarcity [12, 13]. Major Iraqi reservoirs have been investigated, showing considerable surface water extent variations due to drought [11], water regulation by upstream reservoirs, and seasonal hydrological variations, which underscores the need for continuous monitoring of surface water over the reservoirs using geospatial approaches to facilitate sustainable water resource management [16, 17]. SVM and GIS/RS data can provide efficient tools for monitoring environmental changes [18] and evaluating the sustainability indicators of complex ecosystems, such as lakes and wetlands [19, 20].

Hamrin Lake is very important for the provision of water and agricultural resources, fisheries, biodiversity, and ecological balance in the region, and it is considered to be one of the most important freshwater resources in Iraq. The lake has undergone significant environmental changes over the last few decades due to climate variability, extended droughts, a decrease in inflow from tributaries, sedimentation, and increasing human impacts. Water quality and quantity issues have been adversely impacted by these challenges, leading to changes in the lake's surface area and ecosystem degradation. Therefore, monitoring and evaluating these changes in the environment is a necessity to ensure sustainable water resource management and the integrity of the lake's ecosystem.

The novelty of this study does not come from the individual GIS, remote sensing, or machine learning, but rather from their use in combination for Hamrin Lake, a reservoir that has been quantitatively assessed spatiotemporally only a few times. Moreover, the present study presents a multi-temporal assessment (2017–2026) of the lake dynamics based on Sentinel-2 images and SVM classification for sustainable water management. The proposed workflow is an achievable and translatable solution for monitoring Iraqi reservoirs under climatic and hydrological changes.

The aim of this study is to monitor the environmental changes and sustainability status of Hamrin Lake with the use of GIS-based SVM classification in the context of RS information. Multi-temporal satellite imagery is used to study changes in water extent, land use, and vegetation dynamics over the years. The efficacy of the SVM algorithm for the correct classification of environmental characteristics and identification of spatial changes in a lake ecosystem is also evaluated. The obtained results would be useful in the process of sustainable environmental management, as well as for providing scientific data to decision-makers, and the

great importance of using ML together with geospatial technologies in environmental monitoring applications in Iraq.

2. Methodology

2.1 Study area

Hamrin Lake is one of the largest man-made reservoirs in the eastern part of Iraq, created by the construction of the Hamrin Dam on the Diyala River [21]. The lake is situated between the governorates of Diyala and Salah al-Din and is a vital source of irrigation water, flood control, fisheries, and water supply [22]. The climate is semi-arid with hot summers, low rainfall, and high evaporation, which have a significant impact on water levels. Many villages, farmland and bare land are found around the lake [23].

Climatic variability and increasing water demand have led to numerous environmental and hydrological changes in Hamrin Lake in recent years. Hamrin Lake is located in Iraq as indicated in the figure 1.

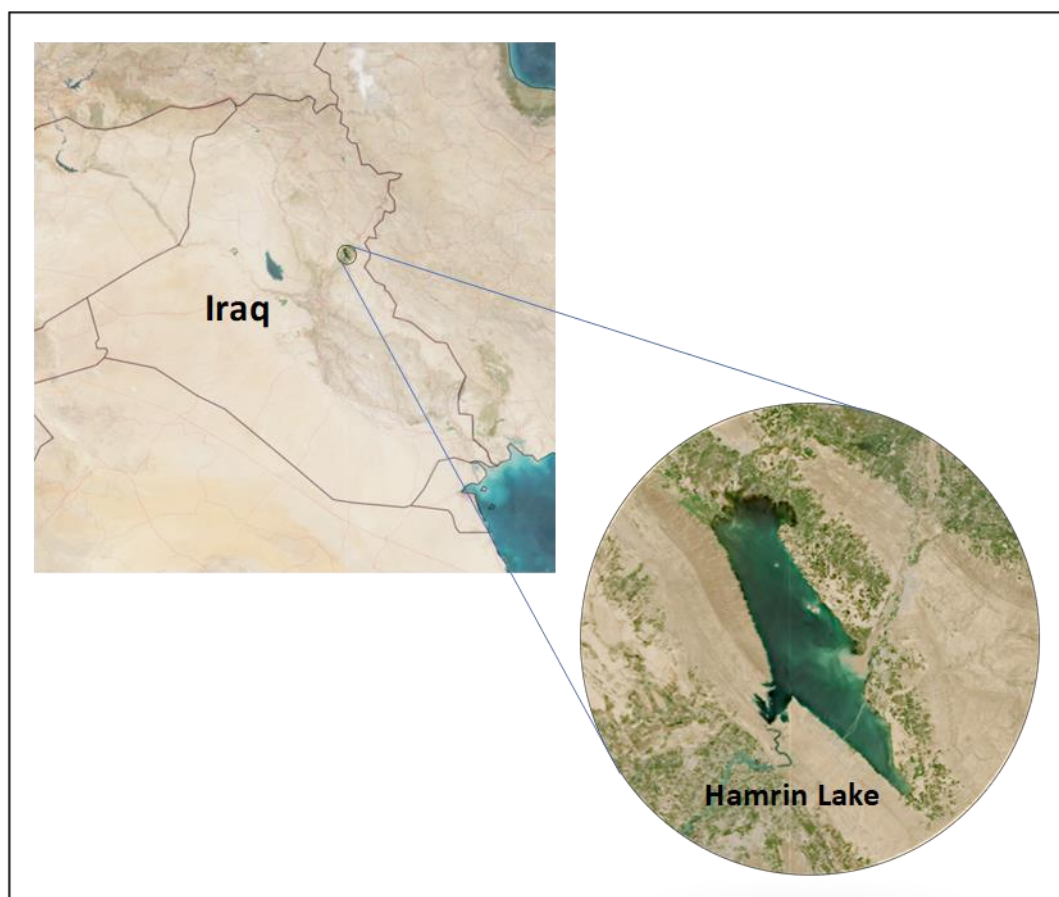


Fig.1. Hamrin Lake's geographical location in Iraq.

2.2 Flowchart

Figure 2 illustrates the general methodology used to monitor the spatial and temporal dynamics of Lake Hamrin. The first step of the study was to download Sentinel-2 Level-2A satellite imagery, along with climatic data such as air temperature and rainfall. The satellite images were processed to ensure spatial consistency by cropping them to fit the study area,

layering them, and projecting them onto the WGS 1984 coordinate system. Representative training samples were then selected from the main land cover classes. The water was classified relative to the surrounding land using the Supporting Vector Machine (SVM) algorithm. These classified images were used to estimate the lake's area and determine its changes between 2017 and 2026. To quantify the spatial and temporal changes, area statistics and thematic maps were prepared. All collected data were integrated into a GIS database for visualization and analysis. Finally, the results were interpreted to determine environmental sustainability and provide decision-making information to aid in the long-term management and conservation planning of Lake Hamrin.

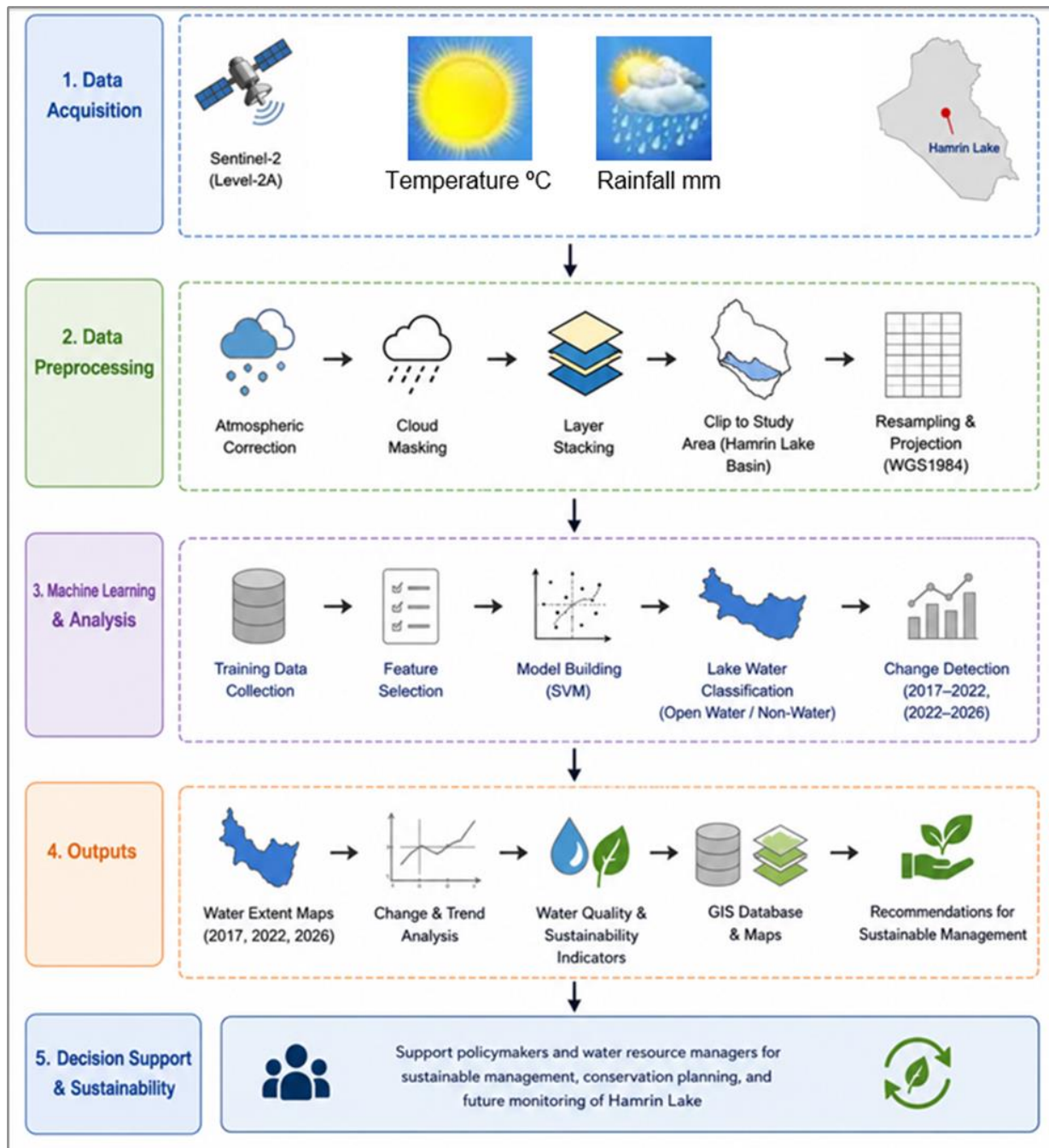


Fig.2. Methodology steps.

2.3 Data acquisition

The first step of the methodology was to gather Sentinel-2 Level-2A satellite imagery to monitor Hamrin Lake. Multi-temporal images were chosen for 2017, 2022, and 2026 to assess temporal changes in the extent of the lake and were downloaded from Copernicus Data Space Ecosystem [24]. Due to their high spatial resolution and free availability for environmental applications, Sentinel-2 imagery was selected. The resolution of the images acquired was 20 m, which is adequate for identifying water boundaries and surrounding land features. The low percentages of cloud coverage enhanced the image quality and reliability of image classification. Images were acquired in the WGS1984 coordinate system to facilitate spatial consistency in analysis. The dates of the selected acquisitions were associated with different hydrological conditions of the lake. The valuable spectral information was provided by satellite data, which was used to distinguish water and non-water classes. The multi-date imagery allowed for the detection of change over time and trend analysis. The major database for the following stage of GIS and ML analysis was created in this stage. Table 1 reports the images' properties. Furthermore, Figure 3 shows annual rainfall and temperature of Hamrin Lake from 2017 to 2026.

Table 1. Images properties.

Image Type	Date of Sensing	Cloud Coverage	Resolution	Coordinate System
Sentinel-2 LA2	10/10/2017	0.0037%	20m	WGS1984
Sentinel-2 LA2	13/04/2022	0.0443%	20m	WGS1984
Sentinel-2 LA2	27/01/2026	0.0095%	20m	WGS1984

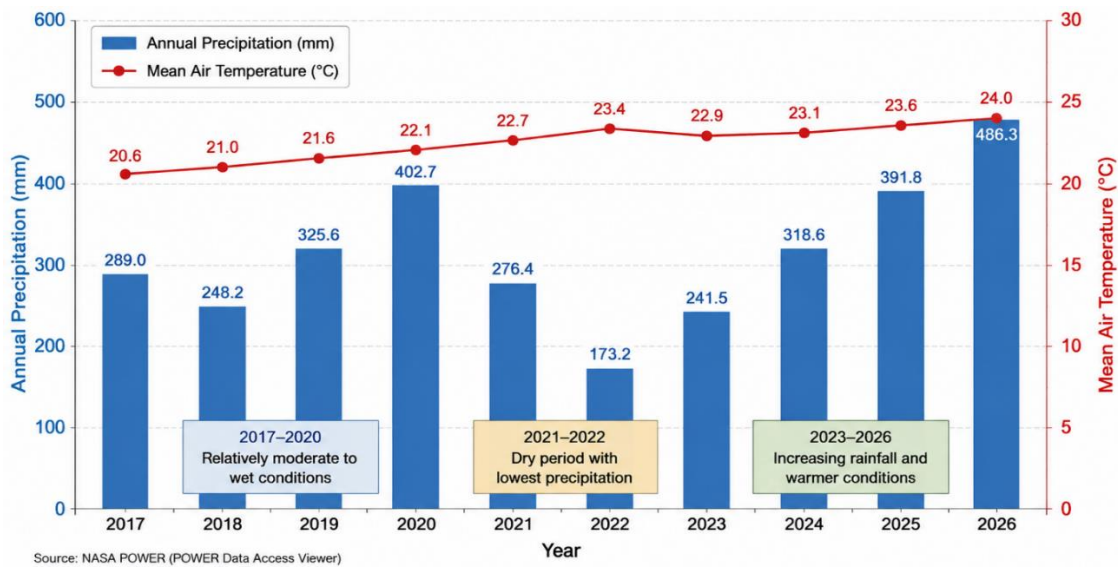


Fig. 3. Annual rainfall and temperature of Hamrin Lake from 2017 to 2026.

Annual precipitation in mm and mean air temperature in °C were downloaded from NASA POWER, a data access viewer online application, on 27th May 2026 [25]. The data show generally drier conditions around and/or about 2022, with less rainfall and greater warming, which agree with the observed decrease in the lake surface area. The recovery in the extent of the lake, on the other hand, is stronger during the years of better precipitation conditions (2026). The results are consistent with the hypothesis that climate variations were a major contributor to the lake level changes recorded, in addition to local water management actions.

2.4 Pre-processing

The image was pre-processed to enhance the image quality and to make the satellite data suitable for classification. The surface reflectance products were used as Sentinel-2 Level-2A (L2A) products where atmospheric correction was already carried out in the Copernicus processing chain. The Scene Classification Layer (SCL) that was delivered with the L2A products was used to mask the cloudy pixels before SVM classification.

Spectral bands from Sentinel-2 were layered to create multi-band composite images after the removal of clouds. To minimize processing time and only process the Hamrin Lake basin, the study area was clipped from the original scenes. All bands were resampled to ensure the same spatial resolution. Spatial compatibility was achieved through projection and coordinate system adjustments, using WGS1984. Water bodies and land features were more visible due to image enhancement. Pre-processing reduced the noise and geometric distortion of the images. These procedures increased the reliability of the ML classification process. The final dataset after processing was suitable for feature extraction and model building.

The image processing and classification were done in ArcGIS_{10.8}. To guarantee that the analysis is spatially consistent, Sentinel-2 Level-2A imagery was projected in the WGS 1984 / UTM Zone 38N coordinate reference system. Optimized training samples for the major land-cover classes were used to implement a Support Vector Machine (SVM) classifier with the default ArcGIS parameters.

2.5 ML and analysis

The machine learning and classification phase was targeted for water body extraction using the SVM classification model. The processed Sentinel-2 images were sampled for training water and non-water classes. The SVM algorithm was trained with these samples to differentiate between land cover classes in terms of their spectra. A series of feature selection methods was applied to select the best spectral bands to detect water.

Samples for training the SVM classifier were visually interpreted from the Sentinel-2 Level-2A images and were manually selected to represent the dominant land-cover classes (water and the surrounding land). The spatial distribution of samples throughout the study area was assessed to evaluate spectral variability and ensure less classification bias. This sampling method provided class signatures that were representative, and the reproducibility and reliability of the classification were enhanced.

The SVM model was then built and optimized with the goal of achieving a higher classification accuracy and minimizing misclassification. The trained model was able to distinguish between open water and the areas around the water.

The two main land-cover classes (Hamrin Lake water versus adjacent land) were manually identified from the training data and implemented into ArcGIS with the SVM classifier. The training samples were selected with visual interpretation of false colour composite images from Sentinel-2, with the aim of having representative spectral signatures. An independent set of validation samples was used to evaluate the classification performance, and a confusion matrix was used to calculate the overall accuracy. The overall accuracy (OA) of approximately 94% demonstrated the accuracy of the Supporting Vector Machine (SVM) classification in detecting the spatial and temporal variations of Lake Hamrin. The lake dynamic classification maps were generated in 2017, 2022, and 2026. The water extent change detection was used to determine water extent differences between the study years. Area changes (increase or decrease) that are

statistically significant were detected. This phase was marked by the success of the use of ML techniques for the environmental monitoring and assessment of lakes' sustainability.

The area of the lake was determined using the SVM-classified raster by extracting the water class and converting it to polygons in ArcGIS. Post-classification, the Raster to Polygon tool was used to create vector polygons to delineate the lake extent for each study year. The surface area was then calculated in the WGS 1984 / UTM Zone 38N coordinate system using the Calculate Geometry Attributes function with the polygon geometry to ensure the area measurement is in square meters. Values calculated were converted to (km²) for comparison. The resulting lake extents were 169.09 km² in 2017, 30.08 km² in 2022, and 185.77 km² in 2026. The values suggest a significant decline through the drought and a vigorous rebound in wetter conditions. All three years were processed using the same projection and consistent processing technique, so the areas calculated were comparable and truly represented changes in the extent of the lakes and not a difference in processing techniques.

2.6 Outputs evaluation

Several thematic maps and analytical results on the sustainability of Hamrin Lake were created at the output stage. Maps of water extent were produced in 2017, 2022, and 2026 to show the spatial distribution changes. These maps clearly show changes in the size of the lake and how much water was available in the lake over time. The temporal change and trends in lake area were quantified through the creation of change and trend analysis graphs. The results were categorized, and water conditions and environmental change indicators were shown to be sustainable.

GIS databases with the spatial layers and attribute data were developed for further analysis and future monitoring. The output produced was accurate in terms of graphical and statistical depiction of lake dynamics. Hydrological changes and environmental stress conditions were facilitated by maps and charts. The products produced can be used for scientific research and environmental management applications. In all the outputs, the outputs were an integrated long-term change of the lake.

2.7 Sustainability evaluation

The final step involved implementing the results for sustainable management and decision-making of the environment. Maps and analytical water products produced in the project will be very useful for the water resource managers and policymakers in Iraq. The results show when water levels have dropped significantly and if there are any potential risks to be worried about for Hamrin Lake. Temporal monitoring and change detection results can be used to develop sustainable management strategies. GIS and RS technologies allow water resources to be continuously monitored at a low cost and high efficiency. The methodology can be used for conservation planning and long-term environmental sustainability assessments. The produced database will enable decision-makers to make better use of water resources and drought management techniques. The study also helps in developing strategies for the future for climate adaptation and ecosystem protection. Sustainable development targets can be achieved through continuous monitoring. Continuous monitoring can help reduce environmental degradation and help towards sustainable development targets.

3. Results

The analysis results from the ML and RS techniques using GIS indicated that the extent of Hamrin Lake surface area (HSA) has experienced a significant change from 2017 to 2026, both in terms of space and time. By using an SVM classification model, the water bodies and land features around the water bodies were identified visually in Sentinel-2 images with a high degree of accuracy and mapped. These variations in the size of the lakes from one year to the next were clearly demonstrated in the classified maps, which represented the impact of climatic variability, water management, and hydrology. The Sentinel-2 image from 2017 showed the lake area of Hamrin Lake to be around 169.09 km² with a relatively stable hydrological status. The lake boundary classification using SVM was proven to be accurate, as it was able to distinguish the lake from barren and agricultural lands, indicating the suitability of the ML methods used for water body extraction. However, the image captured in 2022 showed a huge decrease in the surface area of the lake to almost 30.08 km². The classified map showed very pronounced shrinkage in the southern and central parts of the lake, which appeared to be due to the result of a long drought, decreased inflow of water from the river, and increased evaporation rates. This is a downward trend reflecting the increased environmental stresses and therefore sustainability concerns on water resources in Iraq. An extraordinary increase in the lake area was observed in the year 2026 where the area was 185.77 km². According to the water map based on SVM, the water area has drastically increased compared to 2022, particularly in the northern and eastern parts of the reservoir. This increase may be linked to improved rainfall patterns and/or improved water release arrangements, or seasonal hydrological changes. The results from the three years are compared and show how dynamic Hamrin Lake is and the need for ongoing monitoring. The variation chart also shows the changes in the area of the lake, with the lowest value in 2022 and the highest in 2026. GIS and RS classification worked well for detecting environmental changes, while the SVM classification worked well for the overall sustainability evaluation of Hamrin Lake.

The Overall Accuracy of SVM was found to be around 94%, which was an excellent level of agreement with the reference data. The confusion matrix showed high classification reliability for water and surrounding land classes, and proved the SVM model to be suitable for monitoring the spatiotemporal changes of Hamrin Lake. SVM performs better than other classifiers reported from previous studies in water body classification, e.g., Maximum Likelihood and Random Forest classifiers in water body classification from Sentinel-2 images for binary classification [26, 27].

The results showed the significance of geospatial technologies to facilitate sustainable water resource management and environmental decision-making in Iraq. Figure 4 represents Sentinel-2 image maps of Hamrin Lake. Figure 5 shows SVM-based Hamrin Lake variation from 2017 to 2026. These variations in lake area are shown in a Bar chart in Figure 6.

The spatial variation of Hamrin Lake extent in 2017, 2022, and 2026 is shown in the figure and indicates that the reduction in extent was not uniform across the entire lake. The greatest loss was in the southern and south-eastern part of the reservoir, where areas of shallow water were the first to be lost with lower water availability. By comparison, the water-holding capacity of deeper parts of the central and northern regions was greater, hence better hydrological resilience. The restoration in 2026 was in accord with the original lake morphology, but various areas of the lake were still isolated, indicating that shoreline processes are very responsive to water level changes. The exposed margins in 2022 were found to be

associated with the adjacent barren and agricultural lands, which suggests a direct relationship between the lake contraction and the land use change in the surroundings. As the water level fell, areas that were previously flooded were exposed, thus expanding the dry land mass around the reservoir. In contrast, in 2026, the rewetting of many areas around the lake was caused by the lake expansion, highlighting the close relationship between hydrology and land use. These local observations highlight the need for the combination of GIS and remote sensing in order to delineate vulnerable shoreline areas and to aid reservoir management for sustainable development.

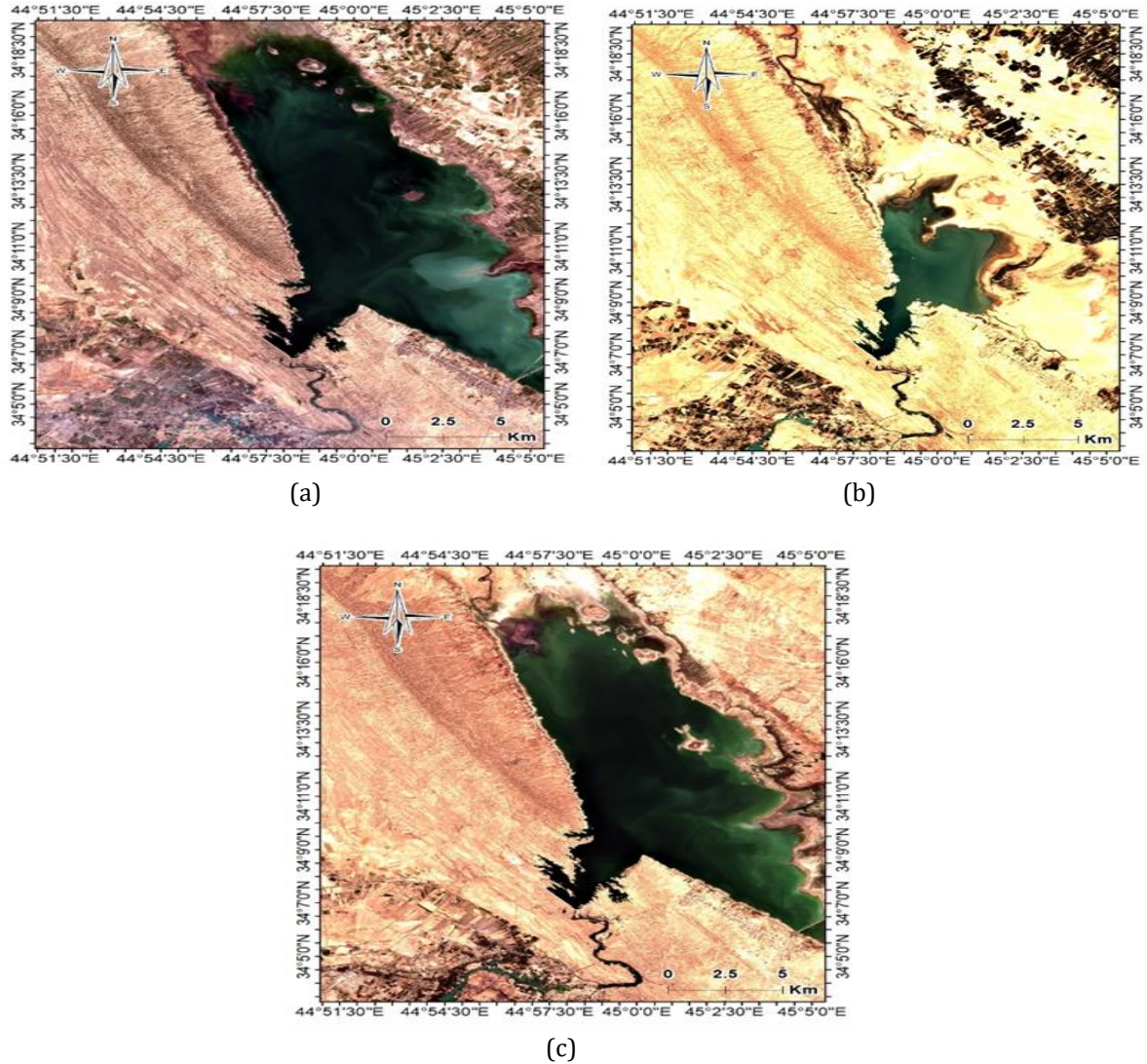


Fig. 4. Sentinel-2 image maps of Hamrin Lake in (a) 2017, (b) 2022, (c) 2026.

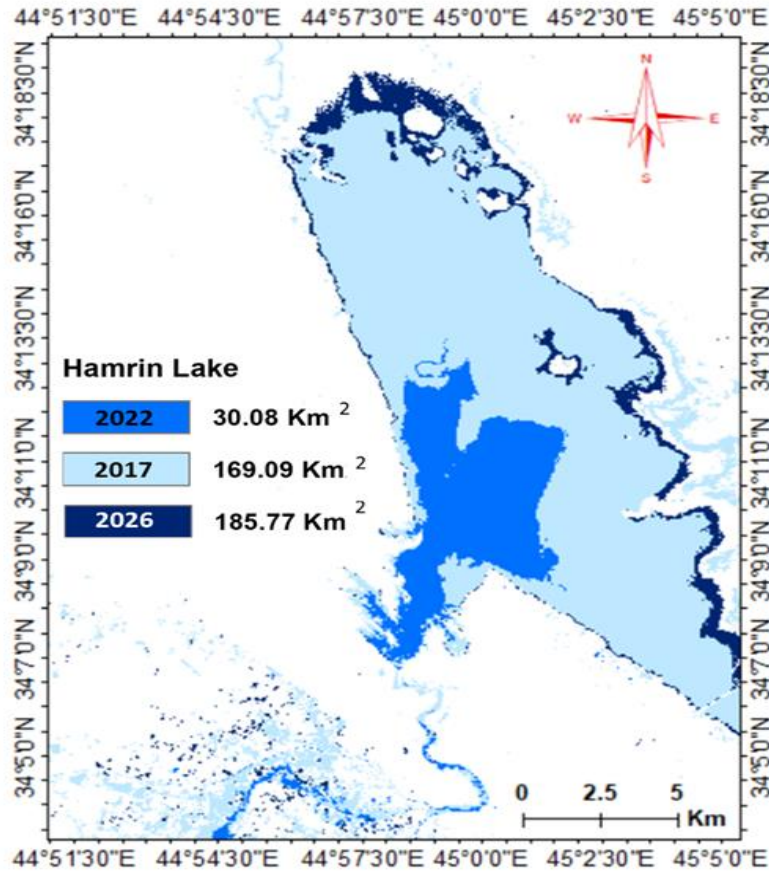


Fig. 5. SVM-based Hamrin Lake variation from 2017 to 2026.

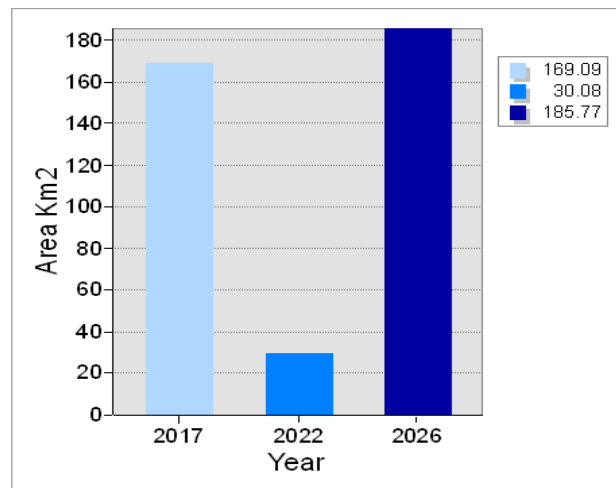


Fig. 6. Bar chart of Hamrin Lake surface water area.

4. Discussion

The results revealed that the water extent of Hamrin Lake during the study period experienced significant variations, which highlights the vulnerability of the water resources in Iraq to climatic and hydrological changes. The extreme decrease in 2022 is indicative of the negative effect of the drought, absence of inflow from tributaries, and high evaporation rates caused by the rise in temperature. The latter, however, shows an improvement in the hydrology in the 2026 expansion, a possible result of more precipitation and water management. The

results of the application of the SVM model are positive and prove that the application of ML techniques using GIS and Sentinel-2 data can be used to monitor the dynamics of lakes. The results are based on the importance of continuous monitoring of water resources through RS for sustainable water resource planning and environmental management in Iraq. Water surface changes around Hamrin Lake are very dynamic in the time series (2017-2026), as indicated by the trend analysis. The lake, in terms of condition, was medium and was hydrologically balanced in 2017, with a relatively stable area of 169.09 km². But in 2022, there was a dramatic decrease in the area of the lake, which sharply decreased to about 30.08 km². This has been a difficult stretch of environmental stress, which has been blamed on a mix of the impacts of years of drought, reduced river flows, increased demand for water, and increased evaporation rates from climate warming. The trend changed in 2026 when the lake area was much larger, approximately 185.77 km², as compared to 2017. The increase is a hydrological recovery that can be attributed to better seasonal inflows, better rainfall and/or better water resource management practices. The overall temporal trend indicates that Hamrin Lake is very sensitive to climatic variability and human use of the water resources. The results also confirmed that the methods of RS, GIS, and SVM have great potential for detecting environmental changes in the long term and to ease the monitoring of the lakes in Iraq in a sustainable way.

5. Conclusions

This study showed the ability of using the integration of GIS, ML methods, and Sentinel-2 RS images to monitor the spatial and temporal dynamics of Hamrin Lake in Iraq, which was confirmed by the results of the study. The lakes were successfully classified with an SVM classification model, and the water bodies and land use around the lakes were demarcated to assess the change in these lakes in the selected years 2017, 2022, and 2026. Thematic maps and statistical analysis clearly showed significant changes in Hamrin Lake extent, which is indicative of its sensitivity to environmental and climatic change. The results showed that the area of the lake in 2017 was 169.09 km² and it reached 30.08 km² in 2022, which is a remarkable decrease. This very low reduction is attributed to the magnitude of the droughts and river flow reductions, increased evaporation and water source stress. Such a reduction on this scale is a significant concern for environmental health, and has the potential to impact water ecosystems, farmers' livelihoods, wildlife, and water safety. The area of the lake in 2026, however, was very impressive, with an area of approximately 185.77 km². This increase is a result of better hydrological conditions, more seasonal variability in precipitation, and/or better water management strategies. Through the trend analysis, it was concluded that there are continuous changes in Hamrin Lake's environment and that the lake should be monitored frequently to ensure sustainable management.

In addition to providing historical knowledge of the lake extent, the study proves that GIS, remote sensing, and machine learning can be used as a viable tool for ongoing environmental monitoring in data-poor areas. The suggested workflow yields reliable spatial information for early detection of environmental changes and water resource planning based on evidence. The application of this is transferable to other Iraqi reservoirs with similar climatic and hydrological conditions. Thus, the study provides a scalable geospatial monitoring scheme that can be used to facilitate sustainable water management and to guide decision-making for environmental protection in the context of a changing climate.

GIS and ML proved to be effective and cost-efficient in detecting environmental changes, generating accurate information in the spatial domain, and supporting decision-making. The proposed method will facilitate the consistent and timely mapping of the dynamics of lakes, which will be useful for the sustainable management of water resources. The present study did not include an economic comparison with conventional monitoring methods, which is recommended for further study.

Furthermore, Sentinel-2 data was shown to be suitable for the monitoring of water resources at high resolution in arid and semi-arid areas. In general, the results of this study emphasize the role of geospatial technologies in attaining environmental sustainability and optimizing water resources management in Iraq. Continuous monitoring coupled with advanced analytical techniques could play a large role in safeguarding lakes and sustainable environmental planning in a changing climate.

This study has a number of limitations that it is important to recognize. Only Sentinel-2 Level-2A data with three representative observation years were used for the analysis and thus do not reflect all conditions in the shorter-term lake dynamics. Furthermore, the level of classification was tested with image-based reference data, instead of large-scale field-based ground truth observations. More studies are needed to combine multi-source satellite data, in situ data, and more frequent temporal observations to further increase the accuracy of monitoring and interpretation.

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